q1_analysis

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Contents

Import and clean dataset

```
pancreatic_df <- read_csv("data/ebl.csv") |>
  janitor::clean_names() |>
  mutate(sex = case_when(sex == 0 ~ "Male",
                         sex == 1 ~ "Female"),
         sex = fct_relevel(sex, "Male"),
         race = case_when(race == 0 ~ "White",
                          race == 1 ~ "Black",
                          race == 2 ~ "Asian",
                          race == 3 ~ "Hispanic"),
         race = fct_relevel(race, "White"),
         op = as.factor(case_when(op == 0 ~ "Body or tail",
                                  op == 1 ~ "Whipple operation")),
         ven_res = as.factor(case_when(ven_res == 0 ~ "No",
                                       ven_res == 1 ~ "Yes")),
         neo_chemo = as.factor(case_when(neo_chemo == 0 ~ "No",
                                         neo_chemo == 1 ~ "Yes")),
         neo_xrt = as.factor(case_when(neo_xrt == 0 ~ "No",
                                       neo_xrt == 1 ~ "Yes")))
```

```
## Rows: 356 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (13): patid, age, sex, race, liver_test, op, ebl, ven_res, neo_chemo, ne...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(pancreatic_df)
```

```
## # A tibble: 6 x 13
##   patid   age sex    race     liver_test op          ebl ven_res neo_chemo neo_xrt
##   <dbl> <dbl> <fct>  <fct>         <dbl> <fct>     <dbl> <fct>   <fct>     <fct>
## 1     1  48.8 Male   White           2.2 Body o~     660 No      No        No
## 2     2  61   Male   White           8.5 Whippl~     800 Yes     Yes       No
```

```
## 3      3  72.3 Female Hispanic        3.5 Body o~   590 No      No         No
## 4      4  58.2 Female White          2.7 Body o~   530 No      No         No
## 5      5  74.8 Male   <NA>           8.9 Body o~   615 No      No         No
## 6      6  36.7 Female Hispanic        3.2 Body o~   680 No      No         No
## # i 3 more variables: les_size <dbl>, or_time <dbl>, prot_energy <dbl>
```

table 1: demographic characteristics + others

```
table1  <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = sex,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",     # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      age ~ "Age (years)",
      race ~ "Race",
      liver_test ~ "Bilirubin Level (in mg/dL)",
      op ~ "Type of Operation",
      ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
      neo_chemo ~ "Chemotherapy Prior to Surgery",
      neo_xrt ~ "Radiation Therapy Prior to Surgery",
      les_size ~ "Tumor Size (in cm)",
      or_time ~ "Operation Time (in mins)"
    )
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Sex**")
table1v2 <- table1 %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width      = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",     # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label       ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
```

```
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()))  |>
  gt::gtsave(filename = "figures_tables/table1.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmpr5jSez/file11f92866a8fd.html screensho
```

```
system("pdfcrop figures_tables/table1.pdf figures_tables/table1.pdf")

table1 |>
  as_kable()
```

Table 1: **Baseline Characteristics by Sex**

Characteristic	**Overall** N = 356	**Male** N = 157	**Female** N = 199	**p-value**
Age (years)	63 (15)	66 (14)	61 (16)	0.011
Race				<0.001
White	217 (73%)	120 (90%)	97 (59%)	
Asian	9 (3.0%)	5 (3.8%)	4 (2.4%)	
Black	22 (7.4%)	3 (2.3%)	19 (12%)	
Hispanic	49 (16%)	5 (3.8%)	44 (27%)	
Unknown	59	24	35	
Bilirubin Level (in mg/dL)	5.3 (3.6)	5.8 (3.9)	4.8 (3.3)	0.020
Type of Operation				0.035
Body or tail	172 (48%)	66 (42%)	106 (53%)	
Whipple operation	184 (52%)	91 (58%)	93 (47%)	
Estimated Intraoperative Blood Loss (in mL)	658 (92)	675 (89)	644 (93)	0.002
Resection + Reconstruction of Major Vasculature	79 (22%)	47 (30%)	32 (16%)	0.002
Chemotherapy Prior to Surgery	67 (19%)	32 (20%)	35 (18%)	0.5
Radiation Therapy Prior to Surgery	39 (11%)	20 (13%)	19 (9.5%)	0.3
Tumor Size (in cm)	3.28 (2.35)	2.86 (1.71)	3.60 (2.71)	0.043
Operation Time (in mins)	374 (134)	404 (140)	351 (125)	<0.001

```
table1byrace  <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = race,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      age ~ "Age (years)",
      sex ~ "Sex",
      liver_test ~ "Bilirubin Level (in mg/dL)",
      op ~ "Type of Operation",
      ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
      ven_res ~ "Resection + Reconstruction of Major Vasculature",
```

```
      neo_chemo ~ "Chemotherapy Prior to Surgery",
      neo_xrt ~ "Radiation Therapy Prior to Surgery",
      les_size ~ "Tumor Size (in cm)",
      or_time ~ "Operation Time (in mins)"
    )
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Sex**")
```

```
## 59 missing rows in the "race" column have been removed.
```

```
table1bv2 <- table1byrace %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width      = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",     # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label      ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()))  |>
  gt::gtsave(filename = "figures_tables/table1b.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmpr5jSez/file11f9758ad461.html screensho
```

```
system("pdfcrop figures_tables/table1b.pdf figures_tables/table1b.pdf")

table1byrace |>
  as_kable()
```

Table 2: **Baseline Characteristics by Sex**

Characteristic	**Overall** N = 297	**White** N = 217	**Asian** N = 9	**Black** N = 22	**Hispanic** N = 49	**p-value**
Age (years)	63 (15)	64 (15)	63 (9)	61 (18)	63 (16)	0.9
Sex						<0.001

Characteristic	**Overall** N = 297	**White** N = 217	**Asian** N = 9	**Black** N = 22	**Hispanic** N = 49	**p-value**
Male	133 (45%)	120 (55%)	5 (56%)	3 (14%)	5 (10%)	
Female	164 (55%)	97 (45%)	4 (44%)	19 (86%)	44 (90%)	
Bilirubin Level (in mg/dL)	5.4 (3.7)	5.8 (3.7)	1.9 (2.1)	5.6 (4.8)	4.0 (2.7)	<0.001
Type of Operation						0.4
Body or tail	144 (48%)	109 (50%)	2 (22%)	11 (50%)	22 (45%)	
Whipple operation	153 (52%)	108 (50%)	7 (78%)	11 (50%)	27 (55%)	
Estimated Intraoperative Blood Loss (in mL)	655 (94)	657 (100)	658 (106)	680 (92)	638 (51)	0.2
Resection + Reconstruction of Major Vasculature	61 (21%)	54 (25%)	2 (22%)	4 (18%)	1 (2.0%)	<0.001
Chemotherapy Prior to Surgery	57 (19%)	52 (24%)	0 (0%)	4 (18%)	1 (2.0%)	<0.001
Radiation Therapy Prior to Surgery	30 (10%)	30 (14%)	0 (0%)	0 (0%)	0 (0%)	0.003
Tumor Size (in cm)	3.29 (2.40)	2.98 (1.75)	2.10 (0.74)	5.22 (3.81)	4.01 (3.49)	0.032
Operation Time (in mins)	373 (134)	378 (145)	386 (102)	418 (50)	328 (100)	0.006

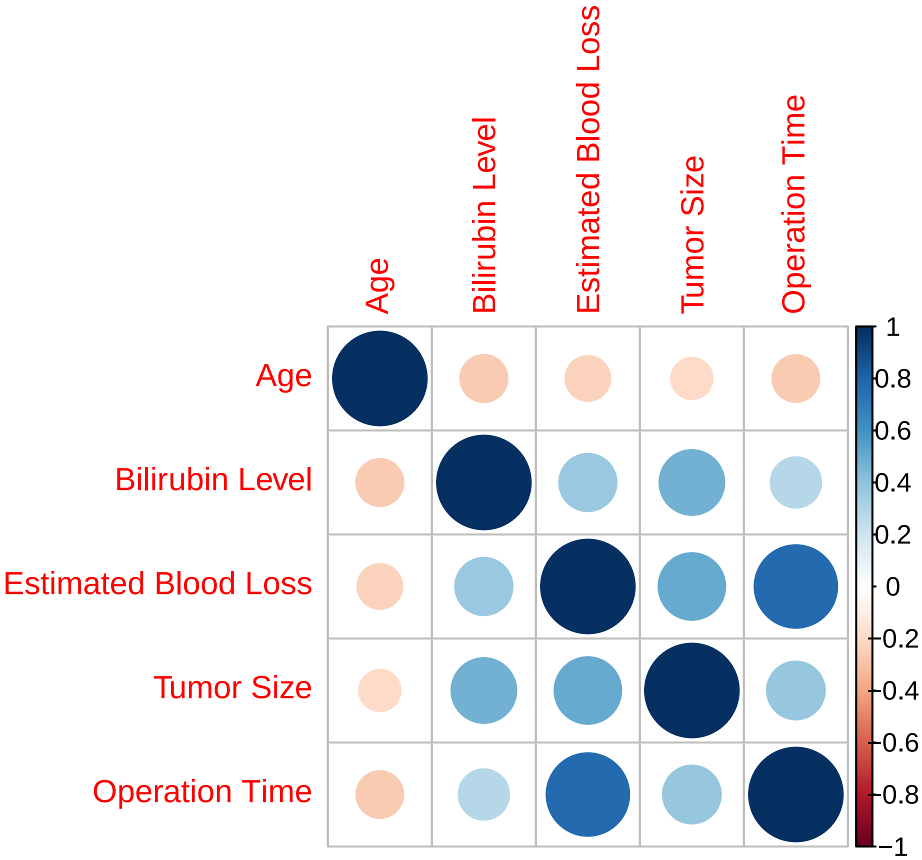
###correlation between predictors and within covariates

Male

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Male") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
pdf("figures_tables/corr_contvars_male.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
##   2
```

```
corrplot(cor_matrix, method = "circle")
```

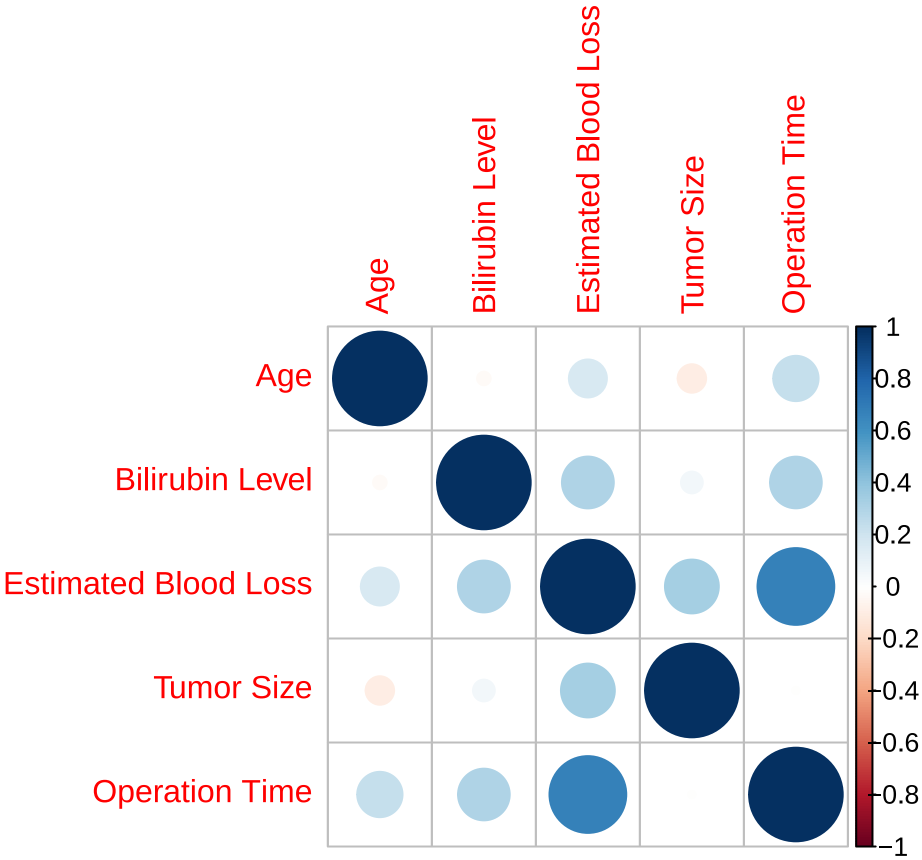


Female

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Female") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Ti
pdf("figures_tables/corr_contvars_female.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
##   2
```

```
corrplot(cor_matrix, method = "circle")
```

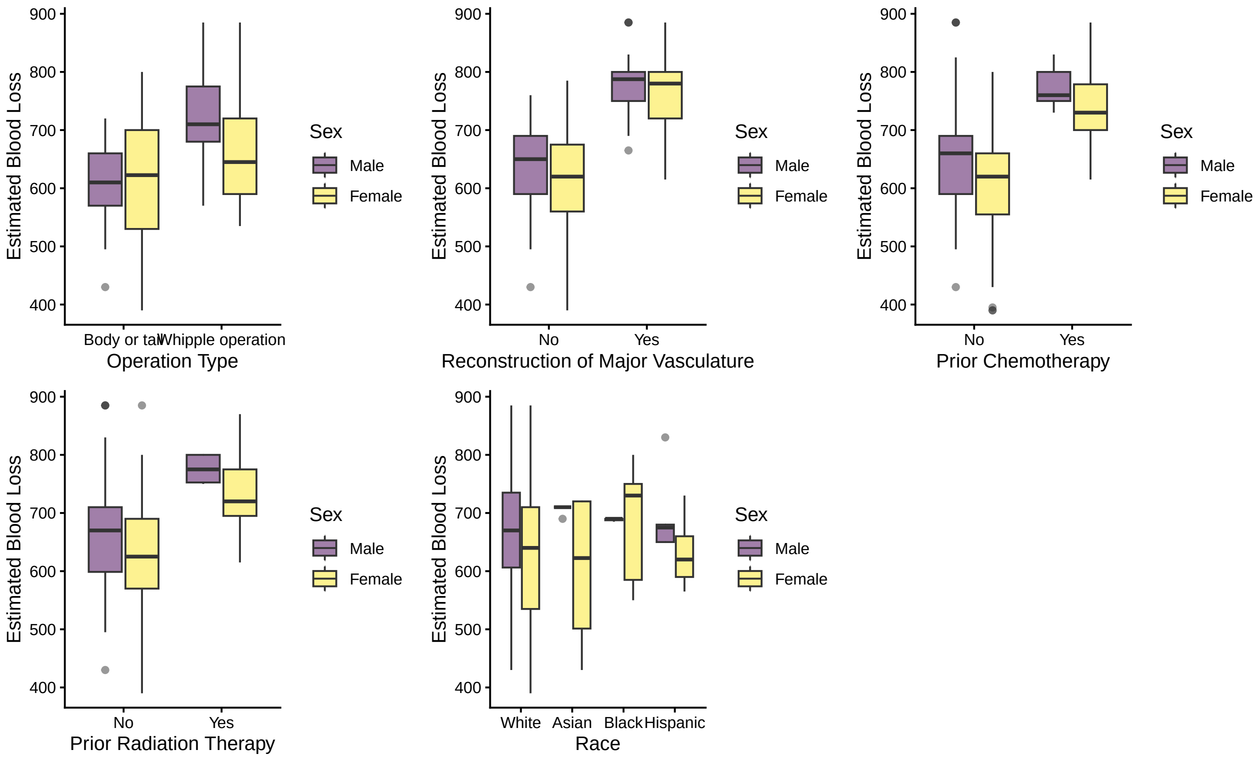


operation time and age negatively correlated in men but no to low positive correlation in women

EBL Distribution in each categorical covariates

```
colnames <- c("op","ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior Radiat
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                 aes(x = get(colnames[x]), y = ebl, fill = sex)) +
    geom_boxplot(alpha = 0.5) +
    labs (x = names[x], y = "Estimated Blood Loss",
          fill = "Sex") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```

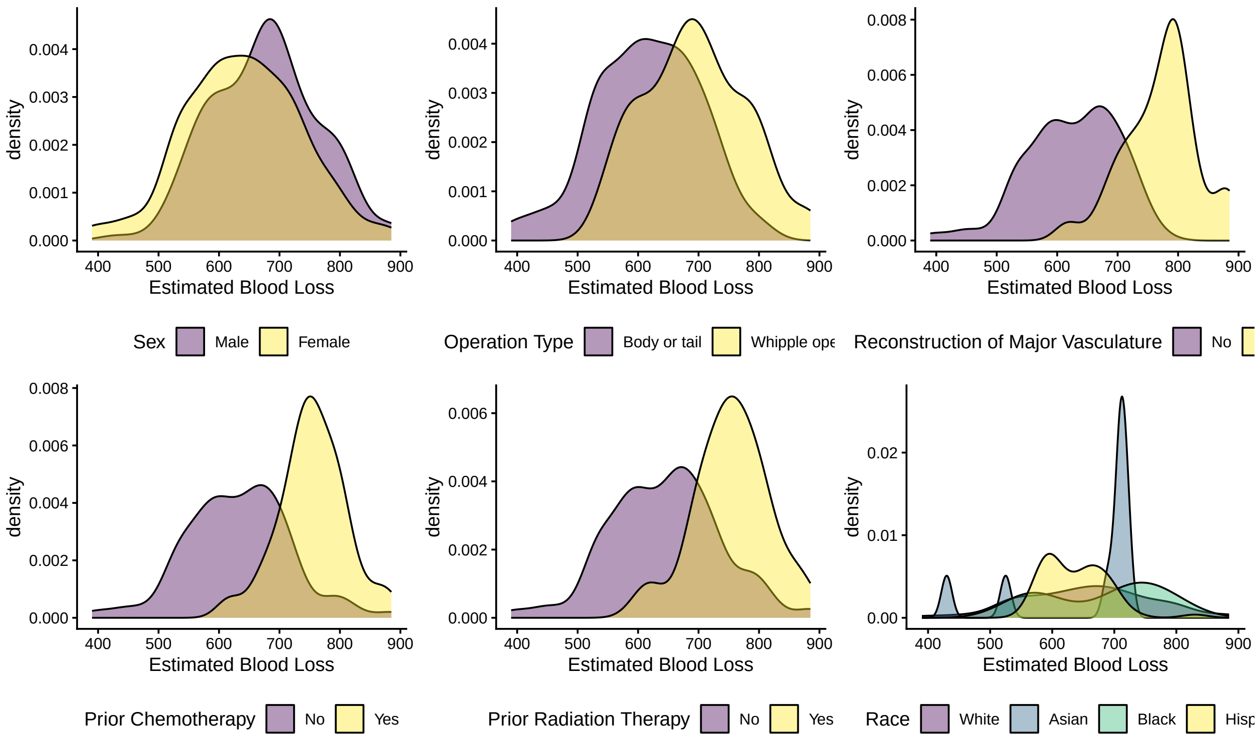



```r
pdf("figures_tables/ebl_dist_across_cat_boxplot.pdf",
       width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
##   2
```

```r
colnames <- c("sex", "op","ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
                 aes(x = ebl, fill = get(colnames[x]))) +
    geom_density(alpha = 0.4) +
    labs (fill = names[x], x = "Estimated Blood Loss") +
    theme_classic() +
    theme(legend.position = "bottom")
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```




```r
pdf("figures_tables/ebl_dist_across_cat_densityplot.pdf",
       width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
##   2
```

EBL lower in female regardless of prior chemotherapy, radiation therapy and major reconstruction status. However, EBL in female vs male are different depending on operation type (effect modification probably, can try interaction term). Low EBL in white and hispanic women. Very few asian and black men recruited in the study so comparison is difficult.

EBL's distribution is relatively normal across each group.

Modeling:

```r
lm_v1 <- lm(ebl ~ race*sex + age +  sex*op + ven_res + neo_chemo +
              neo_xrt + liver_test*les_size + age*or_time,
            data = pancreatic_df)

summary(lm_v1)
```

```
##
## Call:
## lm(formula = ebl ~ race * sex + age + sex * op + ven_res + neo_chemo +
##     neo_xrt + liver_test * les_size + age * or_time, data = pancreatic_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -191.600  -29.133   -2.131   35.287  170.136
##
## Coefficients:
##                                Estimate  Std. Error t value         Pr(>|t|)
## (Intercept)                  346.614950   47.369039   7.317 0.00000000000272
## raceAsian                     61.528157   25.372743   2.425          0.01595
## raceBlack                     50.846418   32.189931   1.580          0.11534
## raceHispanic                 -22.712673   24.988322  -0.909          0.36417
## sexFemale                    -10.063204   10.964294  -0.918          0.35951
## age                            2.381460    0.735324   3.239          0.00135
## opWhipple operation            3.585233   11.625273   0.308          0.75801
## ven_resYes                    25.099429   12.668004   1.981          0.04854
## neo_chemoYes                  61.887432   12.351116   5.011 0.00000096626304
## neo_xrtYes                   -48.124132   15.783700  -3.049          0.00252
## liver_test                    -3.235593    1.861435  -1.738          0.08328
## les_size                       6.394294    2.179019   2.934          0.00362
## or_time                        0.695995    0.120655   5.768 0.00000002128481
## raceAsian:sexFemale         -104.640137   37.477436  -2.792          0.00560
## raceBlack:sexFemale          -85.911010   35.757486  -2.403          0.01693
## raceHispanic:sexFemale        41.086251   26.838920   1.531          0.12694
## sexFemale:opWhipple operation  -10.104737   14.393708  -0.702          0.48325
## liver_test:les_size            1.130844    0.360512   3.137          0.00189
## age:or_time                   -0.005638    0.001858  -3.034          0.00264
##
## (Intercept)                  ***
## raceAsian                    *
## raceBlack
## raceHispanic
## sexFemale
## age                          **
## opWhipple operation
## ven_resYes                   *
## neo_chemoYes                 ***
## neo_xrtYes                   **
## liver_test                   .
## les_size                     **
## or_time                      ***
## raceAsian:sexFemale          **
## raceBlack:sexFemale          *
## raceHispanic:sexFemale
## sexFemale:opWhipple operation
## liver_test:les_size          **
## age:or_time                  **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 53 on 278 degrees of freedom
##   (59 observations deleted due to missingness)
## Multiple R-squared:  0.6998, Adjusted R-squared:  0.6804
## F-statistic: 36.01 on 18 and 278 DF,  p-value: < 0.00000000000000022
```

###Look at missingness

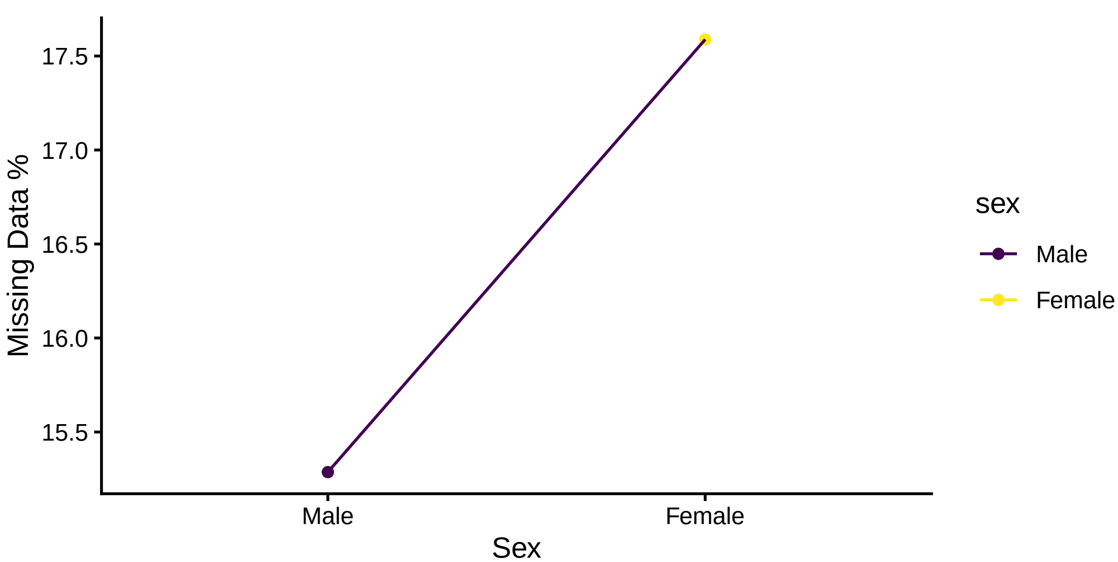
```
fig2b <- vis_miss(pancreatic_df |> dplyr::select(-prot_energy)) +
  theme(axis.text.x = element_text(vjust = 0))
ggsave("figures_tables/missingness_info.png", fig2b, height = 7, width =10)
```

Missingness by gender and age

```
race_gender_missing <- pancreatic_df |>
  dplyr::select(-prot_energy) |>
  group_by(sex) |>
  mutate(total = length(unique(patid))) |>
  ungroup() |>
  drop_na(race) |>
  group_by(sex) |>
  summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
  mutate(dummy_group = 1)

missing_by_gender <- ggplot(aes(x = sex, y = missing_pct, group = dummy_group, color = sex),
       data = race_gender_missing) +
  geom_point() +
  geom_line() +
  geom_line() +
  labs(x = "Sex", y = "Missing Data %") +
  theme_classic()

ggsave("figures_tables/missingness_byracegender.pdf", missing_by_gender, height = 3, width =5)
system("pdfcrop figures_tables/missingness_byracegender.pdf figures_tables/missingness_byracegender.pdf
missing_by_gender
```



Look for other possible sources